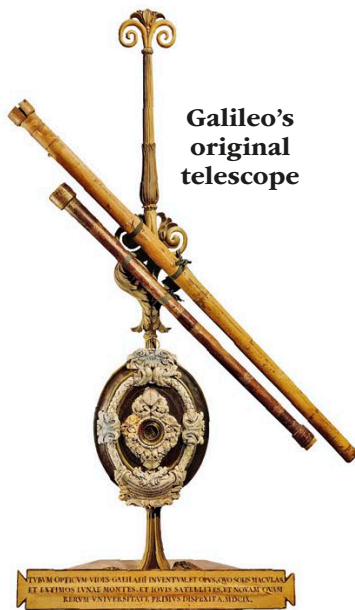




Galileo's
original
telescope

Early telescope

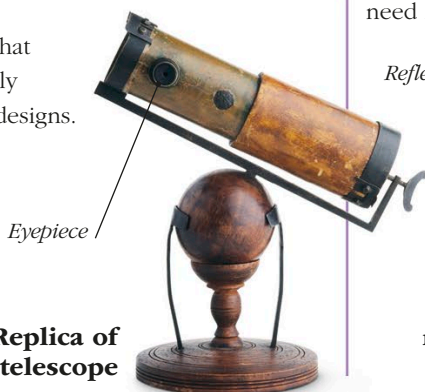
- **What?** Galileo's telescope
- **Who?** Galileo Galilei
- **Where and when?** Italy, 1609

Several astronomers built their own telescopes after hearing of Lippershey's invention, but the most successful was the Italian Galileo Galilei (see pp.258–259). Through careful study, he increased the power of his telescopes from three times magnification (3x) to 20 times (20x), which allowed him to make important discoveries about the universe.

Mirrors for magnification

- **What?** Newton's reflector
- **Who?** Isaac Newton
- **Where and when?** UK, 1668

While studying the properties of light in the 1660s, the English scientist Isaac Newton realized that a curved mirror could bend light to a focus just as well as a lens. His "Newtonian" telescope avoided the distortions that plagued early lens-based designs.



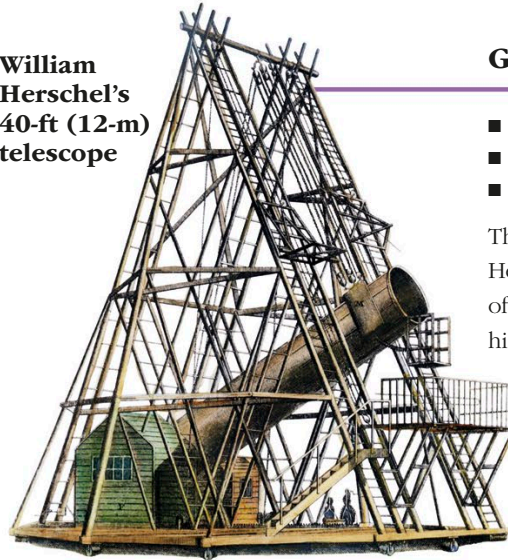
Eyepiece

Replica of
Newton's telescope

Telescopes

Telescopes have come a long way in the four centuries since the lensmaker Hans Lippershey assembled two lenses at either end of a long tube around 1608. Today, they have become huge machines capable of detecting galaxies at the edge of the universe and invisible radiation from gas and dust between the stars.

William Herschel's 40-ft (12-m) telescope



Giant reflectors

- **What?** The first giant telescope
- **Who?** William Herschel
- **Where and when?** UK, 1789

The German-born astronomer William Herschel built the best reflecting telescopes of the time after settling in the UK. After his 1781 discovery of Uranus, he was financed by King George III and built an enormous telescope with a 4-ft (1.2-m) mirror, mounted on a huge turntable that allowed it to point in different directions.

Radio telescope

- **What?** The Lovell Telescope
- **Who?** Bernard Lovell and Charles Husband
- **Where and when?** UK, 1957

The science of radio astronomy began in the 1930s when astronomers discovered radio waves coming from celestial objects other than the sun. Because radio waves are longer than visible light waves, they need much larger telescopes to capture

and focus them. In the 1950s, the British astronomer Bernard Lovell built the first large "dish" telescope at Jodrell Bank near Manchester.

Reflecting metal dish 250ft (76.2m)
across gathers radio waves

Antenna is focal point for
radio waves, which then
generate electrical signals

Lovell Telescope
near Macclesfield,
Cheshire, UK

